

CANVATECH (Anuj Mishra)

Professional Quotation & Cost Breakdown Document

Project: Smart School Management System with Biometric Integration, WhatsApp Automation, Assessment Engine, Payroll System & Mobile Applications

1. Executive Summary

This document presents a comprehensive quotation for the design and development of a production-grade School/Coaching Management Platform. The system includes end-to-end automation covering student lifecycle, teacher operations, biometric attendance integration, real-time notifications, examination workflows, and payroll computation. The solution is engineered using a scalable architecture ensuring high reliability, real-time processing, and seamless integration with external systems such as biometric devices and WhatsApp communication channels.

2. Technology Stack & Architecture

- 1 Backend: Python FastAPI (High-performance asynchronous architecture)
- 2 Database: PostgreSQL (Relational, production-grade)
- 3 Cache & Queue: Redis (Real-time performance & task queue)
- 4 Background Processing: Celery Workers
- 5 Admin Panel: Next.js (React-based modern UI)
- 6 Mobile Apps: Flutter (Android + iOS cross-platform)
- 7 Server: Linux VPS (Dockerized deployment)
- 8 Biometric Integration: eSSL Device (via API / Middleware Sync)
- 9 Messaging System: WhatsApp API with Queue & Retry Mechanism
- 10 Security: JWT Authentication, Role-Based Access Control (RBAC)

3. Detailed Module & Feature Breakdown

3.1 Core Platform Development

- 1 User Authentication System (Admin, Teacher, Student, Parent)
- 2 Role-Based Access Control
- 3 Secure API Architecture
- 4 Session Management & Token Handling
- 5 Centralized Logging & Error Handling

3.2 Student Management System

- 1 Student Profile Creation & Management
- 2 Parent Linking System
- 3 Class / Stream / Section Mapping

- 4 Search & Filtering Mechanism

3.3 Teacher Management System

- 1 Teacher Registration & Approval Workflow
- 2 Section-wise Assignment (Science/Commerce)
- 3 Lecture Assignment System
- 4 Teacher Dashboard Interface

3.4 Attendance System

- 1 Student Attendance Tracking
- 2 Teacher Attendance Tracking
- 3 Daily & Monthly Reports
- 4 Section-wise Attendance Analytics

3.5 Biometric Integration (eSSL)

- 1 Biometric Device Integration
- 2 Punch Log Synchronization Service
- 3 Entry & Exit Detection Logic
- 4 Duplicate Prevention & Validation
- 5 Attendance Event Processing Engine

3.6 WhatsApp Automation System

- 1 Student Entry Notification to Parents
- 2 Exit Notification System
- 3 Teacher Attendance Alerts
- 4 Salary Notification Automation
- 5 Queue-based Message Retry System

3.7 Payroll & Salary Engine

- 1 Late Arrival Detection Logic
- 2 Half-day & Deduction Rules
- 3 Daily Salary Calculation
- 4 Monthly Salary Aggregation
- 5 Downloadable Salary Reports

3.8 Assessment & Examination System

- 1 Test Scheduling System
- 2 Student Test Interface
- 3 Evaluation Logic
- 4 Performance Analytics

3.9 Mobile Application (Flutter)

- 1 Student Mobile Application
- 2 Teacher Mobile Application
- 3 Real-time Notifications
- 4 Attendance & Exam Access

3.10 Admin Panel (Next.js)

- 1 Centralized Dashboard
- 2 Analytics & Reports
- 3 Full System Control Panel
- 4 User & Data Management

3.11 DevOps & Deployment

- 1 Dockerized Application Setup
- 2 Nginx Reverse Proxy Configuration
- 3 SSL (HTTPS) Security Setup
- 4 Server Configuration & Optimization

4. Cost Breakdown

Module	Cost (₹)
Core Platform	25,000
Student Management	15,000
Teacher Management	15,000
Attendance System	18,000
Biometric Integration	30,000
WhatsApp Automation	12,000
Payroll System	18,000
Assessment System	10,000
Mobile Apps	15,000
Admin Panel	7,000
DevOps	3,000

Total Project Cost: ₹1,68,000

5. Terms & Conditions

- 1 Server hosting and domain costs are excluded.
- 2 Google Play & Apple App Store charges are excluded.
- 3 Payment milestones to be mutually agreed.
- 4 Support and maintenance can be provided separately.
- 5 System will be delivered production-ready.